Predicting student final performance with linear regression, decision tree and random forest

University of Technology Sydney

36100 Data Science for Innovation - Spring 2026

Assignment Stage 2

Students:

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1. Literature review

Regression and tree-based models are among the most used techniques in student performance research. Namoun and Alshanqiti (2021) reviewed 62 studies and found regression in 51.61% of them, with neural networks and tree-based models together in 29.02%. Batool et al. (2023) surveyed about 260 studies and report that neural networks and random forests are the most common algorithms, and that academic records and demographic factors are the strongest predictors. Our three datasets cover academic history, study behaviour and demographic background, so one linear model and two tree-based models are a natural set to compare.

Cortez and Silva (2008) compared decision trees, random forests, neural networks and support vector machines on two Portuguese school datasets (395 mathematics and 649 Portuguese-language students), with the final grade G3 as the target. Tree-based methods generally beat the other two, and random forest was the best model in 8 cases against 4 for the decision tree. Prior grades drove accuracy: for mathematics, the random forest RMSE rose from 1.75 with G1 and G2 to 3.90 without them. Ngoc An's dataset has the same structure (395 records, G1 to G3), so we test it with and without G1 and G2. A random forest averages many trees grown on bootstrap samples with random feature subsets, which lowers the variance of a single tree (Breiman, 2001).

Results for the linear model are mixed. Falát and Piscová (2022) predicted GPA for 79 university students. Random forest had the lowest cross-validated error (mean MAPE 11.13%), but a linear regression with backward selection was the most accurate on their 19-student test set, so the evidence rests on a very small sample. Sonkhanani et al. (2026, preprint) used 4,424 students with first-semester grades and demographics. Linear regression reached $R^2 = 0.705$, ahead of random forest (about 0.66 to 0.69) and the tuned decision tree (0.618). Yağcı (2022) classified the final exam results of 1,854 university students into four grade bands. Random forest and a neural network both reached accuracy 0.746, and logistic regression reached 0.717. Cortez and Silva (2008) did not test a linear model, so their results compare tree-based methods and the other learners only. The studies do not agree on a winner, and linear regression does well when a previous grade is among the predictors.

We chose linear regression, decision tree and random forest for three reasons. Linear regression is a transparent baseline. A single decision tree shows which thresholds separate high and low performers. Random forest tests whether averaging many trees improves on both. The literature gives no single winner, so we apply the same three models to three datasets with different predictor types and compare them under one protocol.

2. Experimental setup

Research questions and hypotheses

RQ1: Which of linear regression, decision tree and random forest predicts students' final performance most accurately on each of our three datasets?

RQ2: Does the ranking of the models change with dataset characteristics: the presence of prior grades, the type of predictors and the sample size?

H1: Random forest has the highest test R^2 on most datasets, because it can capture non-linear effects and interactions. It beat decision trees in Cortez and Silva (2008) and had the lowest cross-validated error in Falát and Piscová (2022).

H2: Linear regression is within one standard deviation of the random forest's cross-validated R^2 on Ngoc An's dataset with G1 and G2, where G2 and G3 correlate at r = 0.905.

H3: All three models reach their lowest R^2 on the demographic dataset, which has only categorical predictors.

Datasets and targets

Table 1. Datasets, targets and predictors.

Student	Records	Target	Predictors
Ngoc An Nguyen	395	G3 (0 to 20)	Academic, behavioural, family and social attributes, run with and without G1 and G2
Quang Trung Nguyen	1,000	Final exam score (0 to 100)	Study hours, attendance, sleep hours, internet access, extracurricular activities, part-time job
Akil Venkateshkumar	1,000	Composite score, the mean of math, reading and writing (0 to 100)	Gender, race/ethnicity, parental education, lunch type, test preparation course

Models

All three models are regressors from scikit-learn (Pedregosa et al., 2011). Linear regression (ordinary least squares, standardised numeric features, one-hot encoded categorical features) is the baseline and has no tuning. The decision tree (CART; Breiman et al., 1984) is tuned over max_depth [3, 5, 7, 10, None] and min_samples_leaf [1, 5, 10]. The random forest (Breiman, 2001) is tuned over n_estimators [100, 300], max_depth [5, 10, None] and max_features ["sqrt", 0.5]. Both tree models use random_state=42.

Metrics

R^2 is the main metric because it has no unit. MAE and RMSE are reported in the original score units. G3 runs from 0 to 20 and the other targets from 0 to 100, so we also report RMSE divided by the standard deviation of the target, which allows a comparison across datasets.

Validation

Each student splits the data 80/20 (random_state=42) and tunes the models with 5-fold cross-validation on the training set (Kohavi, 1995), using shuffled folds with random_state=42 and R^2 as the search score. Scaling and encoding sit inside a scikit-learn Pipeline, so they are fitted on the training folds only. The test set is used once, for the final scores. Five folds are too few for a reliable paired test, so we call two models tied when the gap in mean cross-validated R^2 is smaller than the larger of their two fold standard deviations.

3. Implementation

3.1 Student name: Ngoc An Nguyen

Dataset description

Source: https://archive.ics.uci.edu/dataset/320/student+performance

This analysis uses the UCI Student Performance dataset, containing 395 students and 33 variables. The target variable is the final mathematics grade (G3), ranging from 0 to 20. Predictors include academic records, study habits, absences, previous failures, family background and social characteristics.

Two scenarios were investigated: Scenario A includes previous grades (G1 and G2), while Scenario B excludes them. This comparison examines how the availability of prior academic performance affects prediction accuracy.

Preprocessing and feature engineering

Data-quality checks identified no missing values or duplicate observations, so all 395 records were retained. G3 and any target-derived variables were excluded from predictors to prevent data leakage.

Categorical variables were transformed using one-hot encoding, while numerical variables were standardised for Linear Regression. Decision Tree and Random Forest retained their original numerical scales. Preprocessing was implemented within scikit-learn Pipelines to ensure transformations were fitted only on training data.

An 80/20 train-test split (random_state=42) produced 316 training and 79 testing observations. Both scenarios used identical student indices, so their results can be compared directly. Feature engineering focused on categorical encoding rather than creating additional predictors.

Model implementation and tuning

Three regression models were implemented: Linear Regression, Decision Tree and Random Forest. Linear Regression provided an interpretable baseline, Decision Tree captured nonlinear relationships, and Random Forest combined multiple trees to improve predictive performance.

Five-fold shuffled cross-validation was performed on the training data, using R^2 as the optimisation metric. GridSearchCV tuned Decision Tree over max_depth [3, 5, 7, 10, None] and min_samples_leaf [1, 5, 10]. Random Forest was tuned over n_estimators [100, 300], max_depth [5, 10, None] and max_features [sqrt, 0.5]. Both tree models used random_state=42.

For Scenario A, the selected Decision Tree had max_depth=5 and min_samples_leaf=5, while Random Forest used 300 trees, max_depth=10 and max_features=0.5. For Scenario B, Decision Tree selected max_depth=3 and min_samples_leaf=10, while Random Forest used 300 trees, max_depth=5 and max_features=0.5. The optimised models were refitted on the training data before final testing.

3.2 Student name: Quang Trung Nguyen

Dataset

Source:

https://www.kaggle.com/datasets/harshadapatil31/student-performance-and-study-habits-dataset

My dataset is the Kaggle student performance and study habits file: 1,000 students and 12 columns. The target is final_exam_score (mean 83.5, SD 10.3, range 46.8 to 100). A total of 78 students (7.8%) score exactly 100, so the target has a ceiling. Scenario A follows the group protocol and uses six behavioural predictors: study_time_hours, attendance_percent, sleep_hours, internet_access, extracurricular_activities and part_time_job. Scenario B adds previous_grade.

Preprocessing and feature engineering

The file has no missing values, duplicate rows or impossible values, so I removed no rows. Pandas reads the category None in parental_education as missing (102 rows), so I load the file with keep_default_na=False. I dropped student_id and final_grade, a letter band computed from the score that would leak the target. Binary columns are one-hot encoded and numeric columns are scaled for Linear Regression only. Both steps sit inside a scikit-learn Pipeline, so each cross-validation fold fits them on its own training part.

I tested two engineered features: effective_study (study hours x attendance / 100) and sleep_gap (the distance of sleep hours from 7). I set the rule before looking at test scores: keep the new features only if cross-validated R^2 rises by more than one fold SD. The gains were -0.003 for Linear Regression, 0.018 for Decision Tree and 0.006 for Random Forest, against SDs of 0.034, 0.080 and 0.025, so I kept the raw features.

Models and parameter tuning

I split the data 80/20 with random_state=42, giving 800 training and 200 test students. Each model was tuned on the training set with 5-fold shuffled cross-validation scored by R^2, using the grids in Section 2, then refitted on all 800 rows. Linear Regression (ordinary least squares) has no hyperparameters. Table 2 shows the selected values.

Table 2. Selected hyperparameters (5-fold cross-validation on the training set).

Model	Scenario A: behaviour only	Scenario B: with previous_grade
Linear Regression	no hyperparameters	no hyperparameters
Decision Tree	max_depth 5, min_samples_leaf 10	max_depth None, min_samples_leaf 10
Random Forest	300 trees, max_depth 10, max_features sqrt	300 trees, max_depth 10, max_features 0.5

The Decision Tree picked the largest leaf size in the grid (10) in both scenarios, which suggests that stronger regularisation could help. The group protocol fixes the grid, so I did not extend it.

3.3 Student name: Akil Venkateshkumar

Dataset

Source: Kaggle, Students Performance in Exams.

https://www.kaggle.com/datasets/spscientist/students-performance-in-exams

The dataset has 1,000 students and eight columns, with no missing values or duplicate rows. Five columns are categorical: gender, race/ethnicity (groups A to E), parental education level (six categories), lunch type (standard or free/reduced) and test preparation course (none or completed). The other three are the mathematics, reading and writing scores (0 to 100). The target is the composite score, the mean of the three, with a mean of 67.77, a standard deviation of 14.26 and a range of 9 to 100. In the exploratory analysis, the mean composite score rises with parental education, from 63.1 for high school graduates to 73.6 for a master's degree, and is higher for students with a standard lunch, so the predictors carry some signal.

Preprocessing and feature engineering

The three component scores are removed from the predictors because they would reveal the target. The five categorical features are one-hot encoded inside a scikit-learn Pipeline that ignores unseen levels, so the encoder is fitted on the training folds only. Scaling is not needed because there are no numeric predictors. No new features were created, because the group protocol fixes the predictor set. The data are split 80/20 with random_state=42, which gives 800 training and 200 test students. All three models use the same preprocessing Pipeline, and the tree models use random_state=42 so that the results can be reproduced.

Models and parameter tuning

Linear regression (ordinary least squares) is the untuned baseline. The decision tree and random forest were tuned with GridSearchCV, using shuffled 5-fold cross-validation on the training set (random_state=42) with R^2 as the score. The test set was used once, for the final scores. Table 3 lists the grids and the selected values. The selected tree is shallow and the forest has a limited depth, which suggests that deeper trees mostly fit noise when there are only five categorical inputs.

Table 3. Tuning grids and selected hyperparameters.

Model	Search grid	Selected
Decision tree	max_depth [3, 5, 7, 10, None]; min_samples_leaf [1, 5, 10]	max_depth=3, min_samples_leaf=10
Random forest	n_estimators [100, 300]; max_depth [5, 10, None]; max_features [sqrt, 0.5]	n_estimators=300, max_depth=5, max_features=sqrt

4. Results and comparison

4.1 Student name: Ngoc An Nguyen

Model performance

Models were evaluated using cross-validated R^2, test R^2, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and normalised RMSE (RMSE divided by the test target's standard deviation). Higher R^2 and lower errors indicate better predictive performance.

Table 4. Cross-validation and test performance of three regression models under two prediction scenarios.

Scenario	Model	CV R^2 (mean ± SD)	Test R^2	Test MAE	Test RMSE	Test NRMSE
A: with G1/G2	Linear Regression	0.804 ± 0.035	0.724	1.647	2.378	0.522
A: with G1/G2	Decision Tree	0.852 ± 0.059	0.704	1.413	2.462	0.540
A: with G1/G2	Random Forest	0.883 ± 0.036	0.829	1.135	1.873	0.411
B: without G1/G2	Linear Regression	−0.092 ± 0.250	0.141	3.395	4.196	0.921
B: without G1/G2	Decision Tree	0.128 ± 0.207	0.058	3.541	4.394	0.964
B: without G1/G2	Random Forest	0.232 ± 0.097	0.242	3.188	3.943	0.865

Scenario A: with previous grades

Random Forest achieved the best test performance, with R^2=0.829, MAE=1.135 and RMSE=1.873. Its cross-validated R^2 was 0.883 ± 0.036, compared with 0.852 ± 0.059 for Decision Tree and 0.804 ± 0.035 for Linear Regression.

Decision Tree achieved a slightly higher cross-validated R^2 than Linear Regression but a lower test R^2 (0.704 versus 0.724). Although Random Forest and Decision Tree were approximately tied under the group's one-standard-deviation cross-validation rule, Random Forest achieved clearly better test results.

Scenario B: without previous grades

Removing G1 and G2 substantially reduced performance. Random Forest remained the strongest model, achieving the test R^2=0.242, MAE=3.188 and RMSE=3.943. Linear Regression achieved test R^2=0.141, while Decision Tree achieved 0.058.

Random Forest's cross-validated R^2 decreased from 0.883 to 0.232, which shows the importance of earlier academic performance. Linear Regression recorded a negative mean cross-validated R^2 (−0.092), so it generalised poorly without previous grades. Random Forest and Decision Tree were also approximately tied under the group's cross-validation rule, despite different test results.

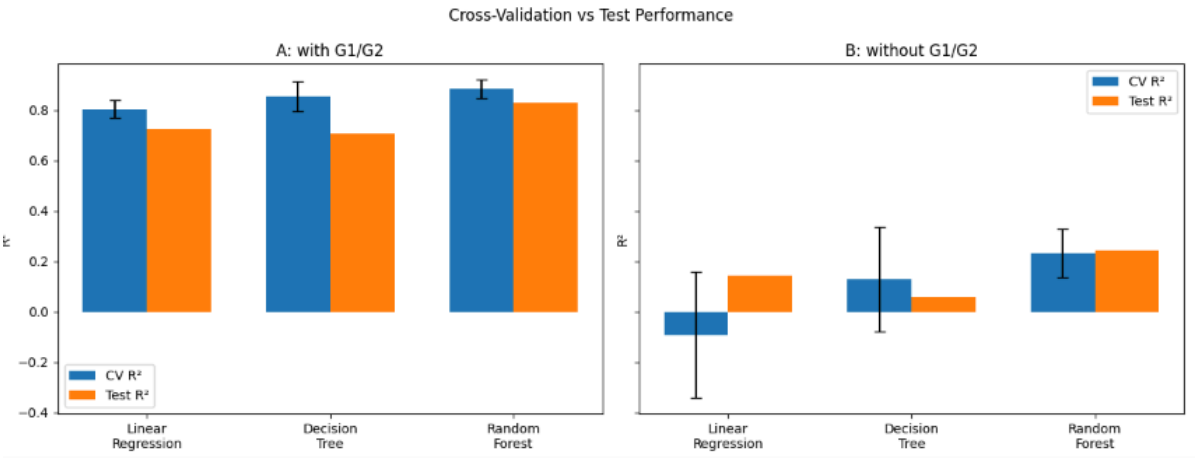


Figure 1. Cross-validation and test R^2 across both scenarios, with error bars representing one standard deviation.

Feature importance and interpretation

Random Forest permutation importance identified G2 as the strongest predictor in Scenario A, followed by G1 and absences. This supports Stage 1 findings that prior academic achievement is strongly associated with final grades.

Without previous grades, failures and absences became the most important predictors, followed by school support and maternal education. However, these variables provided substantially weaker predictive power.

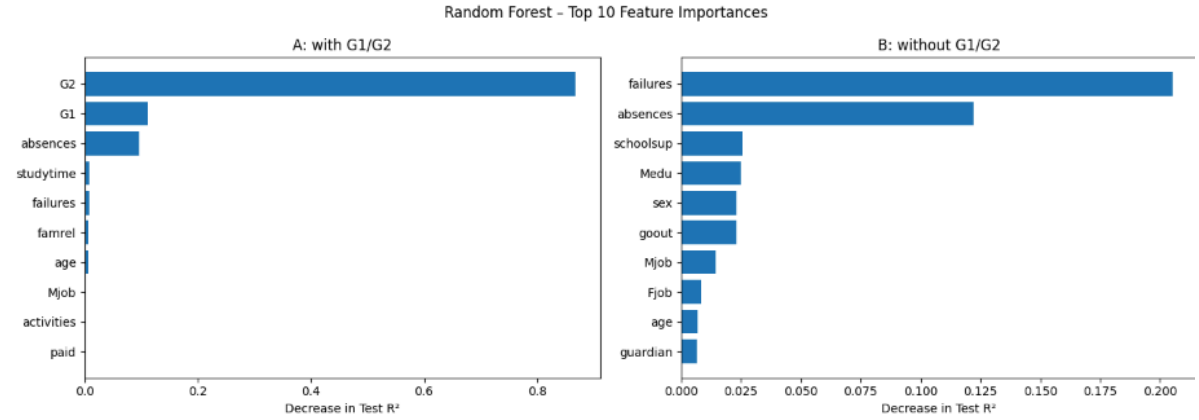


Figure 2. Random Forest permutation features importance across both scenarios.

Best model and conclusion

Random Forest was the best-performing model for this dataset based on test R^2, MAE and RMSE. With G1 and G2, it explained approximately 82.9% of test-set variation in G3, compared with only 24.2% without these grades.

Therefore, Random Forest is recommended when previous academic results are available. Predictions relying only on behavioural, family and background characteristics should be interpreted cautiously because of their limited accuracy. These findings describe predictive relationships rather than causal effects.

4.2 Student name: Quang Trung Nguyen

Results

Table 5 and Figure 3 show the results. MAE and RMSE are in exam points, and NRMSE is RMSE divided by the target SD (10.34).

Table 5. Cross-validated (training set) and test results of the three models.

Scenario	Model	CV R^2 (mean ± SD)	Test R^2	Test MAE	Test RMSE	Test NRMSE
A	Linear Regression	0.480 ± 0.032	0.386	6.58	8.13	0.786
A	Decision Tree	0.328 ± 0.080	0.269	7.23	8.87	0.858
A	Random Forest	0.443 ± 0.025	0.338	6.80	8.44	0.816
B	Linear Regression	0.659 ± 0.047	0.619	4.97	6.40	0.619
B	Decision Tree	0.446 ± 0.065	0.492	5.75	7.39	0.715
B	Random Forest	0.594 ± 0.055	0.585	5.24	6.68	0.646

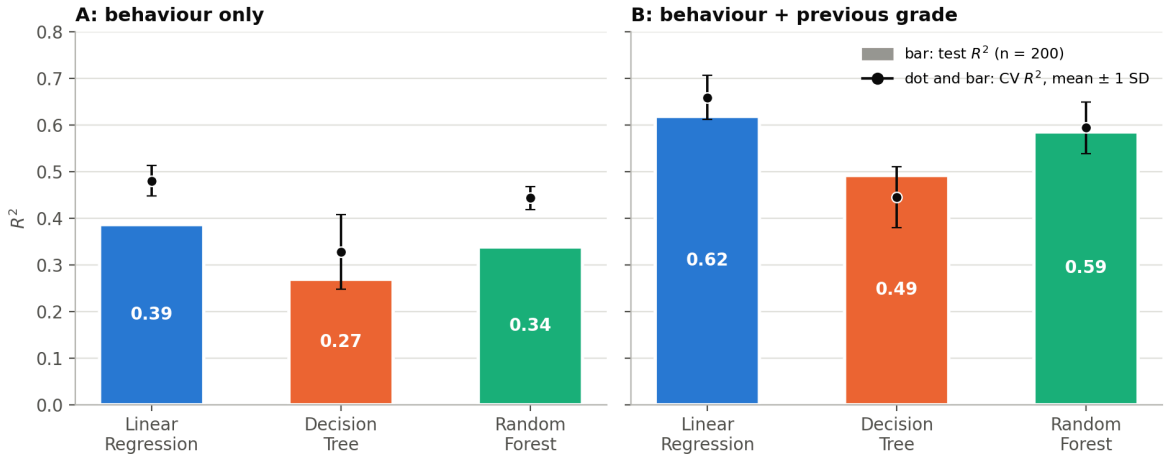


Figure 3. Test R^2 (bars) and cross-validated R^2 (dots, mean ± 1 SD). A: behaviour only. B: with previous_grade.

Comparison and best model

Linear Regression has the highest R^2 in both scenarios. With behavioural predictors only, its test R^2 is 0.386 (cross-validated 0.480 ± 0.032), against 0.338 for Random Forest and 0.269 for Decision Tree. Adding previous_grade lifts the test R^2 to 0.619, 0.585 and 0.492, and cuts the Linear Regression test MAE from 6.58 to 4.97 points. Behaviour alone explains less than half of the variance in exam scores.

The lead needs careful reading. Linear Regression leads Random Forest by 0.037 (A) and 0.065 (B) in cross-validated R^2. The larger fold SD of the two models is 0.032 and 0.055, so the lead sits just above the tie threshold and is modest. Its lead over Decision Tree (0.15 and 0.21) is clear. Across 20 random 80/20 splits, Linear Regression had the highest test R^2 in 19 splits (A) and all 20 (B). My seed-42 test set is harder than average in scenario A (0.386 against a 20-split mean of 0.462), so the cross-validated figures are the better guide.

Random Forest overfits: its training R^2 is 0.84 in scenario A and its test R^2 is 0.34, while Linear Regression moves from 0.49 to 0.39. Linear Regression winning suggests that the effects are close to additive. Its coefficients give +5.85 points per SD of study hours, +3.13 per SD of attendance, +5.74

for internet access and −4.00 for a part-time job. Both models rank study_time_hours first. They disagree after that: attendance is second for the forest (0.16) but fourth by coefficient size in the linear model, and internet_access and part_time_job have large coefficients (+5.74 and −4.00) but small forest importance (0.04 and 0.01).

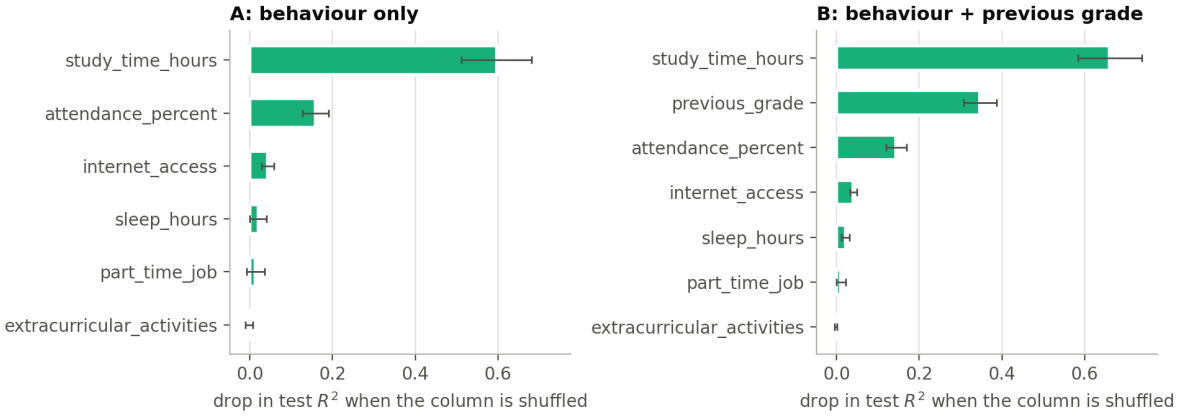


Figure 4. Random Forest permutation importance on the test set (30 repeats).

Best model for my dataset: Linear Regression. It has the highest R^2 and the lowest MAE and RMSE in both scenarios, the smallest gap between training and test R^2, and coefficients that can be read directly. Random Forest is a close second once previous_grade is included (0.585 against 0.619). Capping the few Linear Regression predictions above 100 (2 in A, 6 in B) lifts the scenario B test R^2 to 0.629 and leaves the ranking unchanged. For the group, H1 (Random Forest is best on most datasets) does not hold for my dataset. These are associations from one dataset and a 200-student test set.

4.3 Student name: Akil Venkateshkumar

Results

Table 6 gives the cross-validated R^2 (mean ± SD across five folds) and the test-set metrics. MAE and RMSE are in score points, and RMSE / SD is RMSE divided by the standard deviation of the test target.

Table 6. Model performance on the 200-student test set.

Model	CV R^2 (mean ± SD)	Test R^2	MAE	RMSE	RMSE / SD
Linear regression	0.228 ± 0.037	0.162	10.49	13.40	0.913
Decision tree	0.157 ± 0.060	0.068	10.97	14.14	0.963
Random forest	0.194 ± 0.033	0.118	10.72	13.75	0.937

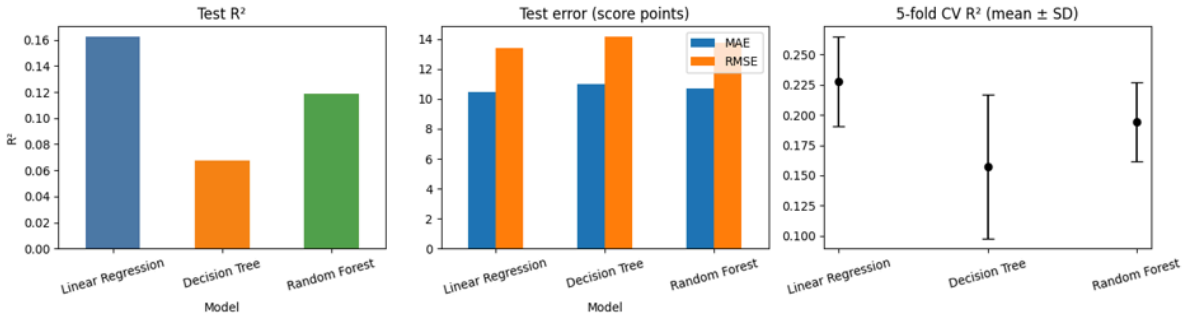


Figure 5. Test R^2, test error and cross-validated R^2 by model.

Comparison and best model

Linear regression has the highest test R^2 (0.162) and the highest cross-validated R^2 (0.228 ± 0.037), followed by the random forest (0.118; 0.194 ± 0.033) and the decision tree (0.068; 0.157 ± 0.060). MAE, RMSE and RMSE / SD give the same ranking. Under the group rule, the random forest ties with linear regression, because its cross-validated R^2 is 0.91 SD lower, within one SD. The decision tree is 1.91 SD lower and is clearly the weakest.

All three models explain a small part of the variance. RMSE is 91% to 96% of the target standard deviation, and the mean absolute error is about 10.5 to 11 points. Test R^2 is lower than cross-validated R^2 for every model, which is consistent with the variation of a single 200-student split. The decision tree is the least stable across folds (SD 0.060, against 0.033 for the forest and 0.037 for linear regression). This supports the point that averaging trees reduces variance (Breiman, 2001), but the forest still could not beat the linear model.

The forest and the linear model point to the same drivers. Lunch type (0.175 and 0.160 for its two levels) and test preparation (0.134 and 0.128) are the most important features in the forest. In the linear model, a standard lunch adds 4.6 points compared with free/reduced lunch, a completed preparation course adds 3.9 points, and parental education shifts the score from −4.1 for a high school diploma to +4.1 for a bachelor's degree.

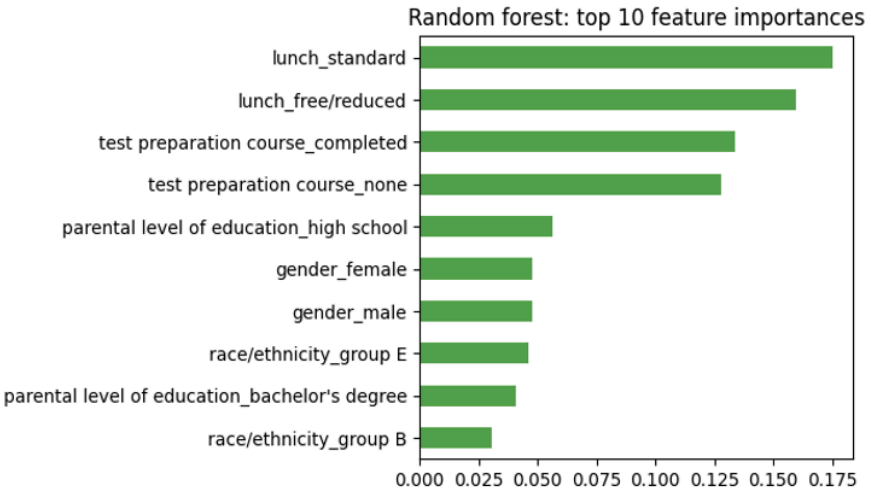


Figure 6. Random forest feature importances (top ten one-hot levels).

Linear regression is the best model for this dataset. It has the highest test and cross-validated scores, needs no tuning and is the easiest to interpret. The effects look mostly additive, so the interactions that trees can find add variance without improving accuracy on 800 training students. The random forest is tied under the group rule, so the choice rests on simplicity. The best test R^2 is only 0.162: these factors separate groups of students but cannot predict an individual student's score accurately. This dataset does not support H1, because the random forest is not ranked first, and its low R^2 is in line with H3.

In practice, linear regression is the right choice when an interpretable model is needed. The random forest could help if more predictors or interactions are added later, but with these five attributes it gives no advantage. The single decision tree shows thresholds clearly, but its test R^2 of 0.068 is too low to rely on.

5. Group comparison and conclusion

No model is best on every dataset. Linear regression wins on the two 1,000-student datasets, random forest wins on Ngoc An's data, and the decision tree is last in all four settings (Table 7).

Table 7. Test R^2 of each model, with the cross-validated R^2 (mean ± SD) in brackets.

Dataset setting	Records	Linear regression	Decision tree	Random forest	Best model
Ngoc An, with G1 and G2	395	0.724 (0.804 ± 0.035)	0.704 (0.852 ± 0.059)	0.829 (0.883 ± 0.036)	Random forest (tied with decision tree)
Ngoc An, without G1 and G2	395	0.141 (−0.092 ± 0.250)	0.058 (0.128 ± 0.207)	0.242 (0.232 ± 0.097)	Random forest (tied with decision tree)
Quang Trung, behaviour only	1,000	0.386 (0.480 ± 0.032)	0.269 (0.328 ± 0.080)	0.338 (0.443 ± 0.025)	Linear regression
Akil, demographics	1,000	0.162 (0.228 ± 0.037)	0.068 (0.157 ± 0.060)	0.118 (0.194 ± 0.033)	Linear regression (tied with random forest)
Mean rank of test R^2		**1.5**	**3.0**	**1.5**	**Linear regression and random forest**

Best model overall

Linear regression and random forest tie on mean rank of test R^2 across the four settings (1.5 each). Adding Quang Trung's previous_grade setting (linear regression 0.619, random forest 0.585, decision tree 0.492) gives mean ranks of 1.4, 1.6 and 3.0. Linear regression also wins on two of the three datasets, so we name it the best model overall by a small margin. Random forest wins only on Ngoc An's data, where it ties with the decision tree under the one-SD rule.

Dependence on dataset characteristics

Prior grades change accuracy more than the choice of model. Random forest falls from 0.829 to 0.242 without G1 and G2, and linear regression rises from 0.386 to 0.619 with previous_grade. The best model's NRMSE shows the same order: 0.41 for Ngoc An with G1 and G2, 0.79 for Quang Trung, 0.87 for Ngoc An without G1 and G2, and 0.91 for Akil. Random forest wins on Ngoc An's 33 mixed variables. Linear regression wins on the short, mostly additive predictor sets of Quang Trung (six behavioural) and Akil (five categorical). Random forest won on the smallest dataset, so sample size does not explain the ranking.

Hypotheses

H1 is not supported: random forest is best in two of the four settings. H2 is not supported either. With G1 and G2, linear regression's cross-validated R^2 is 0.079 below random forest's, more than twice the fold SD (0.036). H3 is partly supported: Akil's dataset is the lowest for random forest (0.118), but Ngoc An's setting without G1 and G2 is lower for linear regression (0.141) and the decision tree (0.058).

Recommendation

Linear regression should be the default model: it needs no tuning and its coefficients show the size of each effect. Random forest suits data with prior grades and many mixed predictors, if its training and test R^2 are compared to catch overfitting. The decision tree is best kept for showing thresholds to

teachers, since it ranks last for prediction. These results come from one 80/20 split per dataset (79 or 200 test students) and show associations only.

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